INCH POUND
MIL-DTL-24127A(SH)
w/AMENDMENT 1
19 March 2019
SUPERSEDING
MIL-DTL-24127A(SH)
29 November 2016

# DETAIL SPECIFICATION

# EDUCTORS, GASOLINE, OIL, OR WATER

DELETED

This specification is approved for use by Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers eductors for seawater, freshwater, oil, or gasoline service on board naval ships.

1.2 Classification. Eductors are of the following types, as specified (see 6.2).

1.2.1 Types.

a. Type A – Stationary.

b. Type B – Portable.

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-H28 - Screw-Thread Standards for Federal Services

COMMERCIAL ITEM DESCRIPTIONS

A-A-59227 - Coupling Assembly, Hose, Fire Fighting

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 4420

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DEPARTMENT OF DEFENSE SPECIFICATIONS

| | | |
|---|---|---|
| MIL-DTL-1183 | - | Fittings, Pipe, Cast Bronze, Silver Brazing, General Specification for |
| MIL-A-8625 | - | Anodic Coatings for Aluminum and Aluminum Alloys |
| MIL-DTL-12165 | - | Strainers, Suction, Firehose; and Strainers, Suction, Hose |
| MIL-DTL-15024 | - | Plates and Tags for Identification of Equipment, General Specification for |
| MIL-C-15726 | - | Copper-Nickel Alloy, Sheet, Plate, Strip, Bar, Rod, and Wire |
| MIL-PRF-20042 | - | Flanges, Pipe, Bronze (Silver Brazing) |
| MIL-C-24723 | - | Castings, Nickel-Copper Alloy |
| MIL-DTL-52404 | - | Connections: Hose, Fire and Water |

DEPARTMENT OF DEFENSE STANDARDS

| | | |
|---|---|---|
| MIL-STD-1916 | - | DoD Preferred Methods for Acceptance of Product |

(Copies of these documents are available online at http://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

| | | |
|---|---|---|
| 803-8436633 | - | Eductor, Portable, Peripheral Jet |

(Copies of this document are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Copies of this document may also be obtained from the Naval Ships Engineering Drawing Repository (NSEDR) online at https://199.208.213.105/webjedmics/index.jsp. To request an NSEDR account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@navy.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

| | | |
|---|---|---|
| ASTM A494/A494M | - | Standard Specification for Castings, Nickel and Nickel Alloy |
| ASTM A582/A582M | - | Standard Specification for Free-Machining Stainless Steel Bars |
| ASTM B584 | - | Standard Specification for Copper Alloy Sand Castings for General Applications |

(Copies of these documents are available online at www.astm.org.)

SAE INTERNATIONAL

| | | |
|---|---|---|
| SAE-AMS-5513 | - | Steel, Corrosion-Resistant, Sheet, Strip, and Plate, 19Cr - 9.2Ni (SAE 30304), Solution Heat Treated |
| SAE-AMS-5516 | - | Steel, Corrosion-Resistant, Sheet, Strip, and Plate, 18Cr - 9.0Ni (SAE 30302), Solution Heat Treated |
| SAE-AMS-5517 | - | Steel, Corrosion Resistant, Sheet and Strip, 18Cr - 8Ni (SAE 301) Cold Rolled, 125 ksi (862 MPa) Tensile Strength |



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| | | |
|---|---|---|
| SAE-AMS-5518 | - | Steel, Corrosion Resistant, Sheet and Strip 18Cr - 8Ni Cold Rolled, 150 ksi (1034 MPa) Tensile Strength |
| SAE-AMS-5519 | - | Steel, Corrosion Resistant, Sheet and Strip, 18Cr - 8Ni (SAE 30301) Cold Rolled, 185 ksi (1276 MPa) Tensile Strength |
| SAE-AMS-5524 | - | Steel, Corrosion and Heat-Resistant, Sheet, Strip and Plate, 18Cr - 13Ni - 2.5Mo (SAE 30316), Solution Heat Treated |
| SAE-AMS-5901 | - | Steel, Corrosion-Resistant, Sheet, Strip, and Plate 18Cr - 8Ni (SAE 30301) Solution Heat Treated |
| SAE-AMS-5902 | - | Steel, Corrosion-Resistant, Sheet and Strip, 18Cr - 8Ni (SAE 30301) Cold Rolled, 3/4 Hard, 175 ksi (1207 MPa) Tensile Strength |
| SAE-AMS-5903 | - | Steel, Corrosion-Resistant, Sheet, Strip, and Plate, 18Cr - 9.0Ni (302), Cold Rolled, 1/4 Hard,125 ksi (862 MPa) Tensile Strength |
| SAE-AMS-5904 | - | Steel, Corrosion-Resistant, Sheet and Strip, 18Cr - 9.0Ni (302), Cold Rolled, 1/2 Hard, 150 ksi (1034 MPa) Tensile Strength |
| SAE-AMS-5905 | - | Steel, Corrosion-Resistant, Sheet and Strip, 18Cr - 9.0Ni (302), Cold Rolled, 3/4 Hard, 175 ksi (1207 MPa) Tensile Strength |
| SAE-AMS-5906 | - | Steel, Corrosion-Resistant, Sheet and Strip, 18Cr - 9.0Ni (SAE 30302), Cold Rolled, Full Hard, 185 ksi (1276 MPa) Tensile Strength |
| SAE-AMS-5907 | - | Steel, Corrosion and Heat-Resistant, Sheet, Strip, and Plate, 17Cr - 12Ni - 2.5Mo (SAE 30316), Cold Rolled, 1/4 Hard, 125 ksi (862 MPa) Tensile Strength |
| SAE-AMS-5910 | - | Steel, Corrosion-Resistant, Sheet, Strip, and Plate, 19Cr - 9.2Ni (304) Cold Rolled, 125 ksi (862 MPa) Tensile Strength |
| SAE-AMS-5911 | - | Steel, Corrosion-Resistant, Sheet and Strip, 19Cr - 9.2Ni (SAE 30304) Cold Rolled, 1/2 Hard, 150 ksi (1034 MPa) Tensile Strength |
| SAE-AMS-5912 | - | Steel, Corrosion-Resistant, Sheet and Strip 19Cr - 9.2Ni (SAE 30304) Cold Rolled, 3/4 Hard, 175 ksi (1207 MPa) Tensile Strength |
| SAE-AMS-5913 | - | Steel, Corrosion-Resistant, Sheet and Strip 19Cr - 9.2Ni (SAE 30304) Cold Rolled, Full Hard, 185 ksi (1276 MPa) Tensile Strength |
| SAE-AMS-P-83461 | - | Packing, Preformed, Petroleum Hydraulic Fluid Resistant, Improved Performance at 275 °F (135 °C) |

(Copies of these documents are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 Component parts. The eductor shall consist of a body (suction chamber), discharge tail (diffuser) with separate or integral throat, internal removable pressure nozzle, and pressure, suction, and discharge connections.



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3.2 Design.

3.2.1 Eductor size. Unless otherwise specified (see 6.2), Type A eductors shall be sized as needed for the intended application, and Type B eductors shall be appropriately selected from the sizes defined in Drawing 803-8436633.

3.2.2 Nonclogging design. The eductors shall be of a nonclogging design, effecting as near as practicable a smooth, streamlined design of the passage from the suction through to discharge. Projection of the throat into the mixing chamber, with consequent dead spaces and eddies where solids may accumulate, shall be avoided.

3.2.3 Nozzle tip. The design of the nozzle tip and the throat of the tail or discharge shall be such that the flow of fluid at the specified capacity will not produce excessive erosion of the parts under service operating conditions as specified (see 6.2).

3.2.4 Nozzle access. The eductor design shall permit easy access to the nozzle for inspection and replacement without removing the eductor from the service line in which it is installed (actuating medium line excepted).

3.2.5 Foot (check) valve and strainer. When specified (see 6.2), the eductor shall be provided with a foot (check) valve and strainer on the suction connection. The strainer shall be either of a cylindrical or flat plate design as specified (see 6.2).

3.2.5.1 Cylindrical strainers. Cylindrical strainers shall have approximately the same length as the diameter and shall be either of cast or fabricated construction. The diameter of the strainer holes shall be not less than ⅛ inch or more than half the diameter of the base of the diffuser throat. The total area of the holes shall be not less than four times the area of the suction inlet.

3.2.5.2 Flat plate strainers. Flat plate strainers shall be designed to withstand a differential pressure of 20 pounds per square inch (psi) without showing any signs of permanent distortion. The total free area of the strainer holes shall be as specified in 3.2.5.1.

3.2.6 End connections. End connections (type, sizes, and pertinent details) shall be as specified (see 6.2) and as follows:

a. Pipe thread connections or threaded parts shall be in accordance with FED-STD-H28.

b. Hose fittings shall be in accordance with A-A-59227, MIL-DTL-52404, and MIL-DTL-12165.

c. Union ends shall be threaded and machined as the thread piece and shall be furnished with ring and tail piece assembled in place in accordance with MIL-DTL-1183.

d. Flanges shall be in accordance with MIL-PRF-20042.

3.3 Marking for identification. Each eductor shall have the following data stamped or cast on the body or contained on a permanently attached nameplate:

a. Manufacturer’s name.

b. Manufacturer’s serial number, model number, and size.

c. Suction lift in feet of water and suction capacity.

d. Discharge head.

e. Driving fluid pressure in psi and volume (gallons per minute [gpm]).

f. Federal standard stock number of the assembly.

g. Federal standard stock number or part number of the nozzle or nozzle tip.

Nameplates, if provided, shall be in accordance with Type A, B, or C of MIL-DTL-15024.



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3.4 Type A eductors. Material for Type A eductors, except the pressure nozzle, shall be bronze in accordance with ASTM B584. Strainers and foot valves shall be of the same material as the eductor. The nozzle may be made in one piece or may have a threaded removable tip. The inner surface of the tip shall form a smooth continuous surface with the inner surface of the nozzle. The one-piece nozzle or the nozzle with tip shall be made of nickel-copper-silicon alloy in accordance with MIL-C-24723 and ASTM A494/A494M. O-rings employed in the interface of component parts of the eductor shall be in accordance with SAE-AMS-P-83461.

3.4.1 Bolting to ship structure. Each eductor shall be provided with lugs or a bracket cast integrally with the body for bolting the eductor to the ship's structure.

3.4.2 Connections. Unless otherwise specified (see 6.2), connections 2 inches and smaller shall be union end and connections 2½ inches and larger shall be flanged.

3.5 Type B eductors. Unless otherwise specified (see 6.2), Type B eductors shall be in accordance with 803-8436633. Where bronze is specified for the eductor, materials of the eductor, strainer, and foot valve shall be the same as for Type A eductors (see 3.4).

3.5.1 Foot valves and strainers. Foot valves for aluminum eductors shall be of copper-nickel or corrosion-resistant steel in accordance with any of the following: MIL-C-15726; SAE-AMS-5513, SAE-AMS-5516 through SAE-AMS-5519, SAE-AMS-5524, SAE-AMS-5901 through SAE-AMS-5907, and SAE-AMS-5910 through SAE-AMS-5913; and ASTM A582/A582M, respectively. Strainers may be of similar material as furnished for the foot valve or of sheet steel that is zinc coated after fabrication.

3.5.2 Aluminum parts. Aluminum parts shall be anodized in accordance with MIL-A-8625 after machining and before assembly.

3.6 Onboard repair parts. Onboard repair parts shall consist of one set of gaskets for each eductor furnished.

3.7 Workmanship. Eductors shall be free of porosity or other casting defects. Threads shall be smooth, properly formed, and free of burrs, nicks, and tears. The eductor shall show no signs of leaking or sweating when tested in accordance with 4.2.6.

3.8 Operating requirements. Operating requirements shall be as specified (see 6.2).

## 4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as conformance inspections (see 4.2).

4.2 Conformance inspection. Conformance inspection shall include the examinations of 4.2.4 and the tests of 4.2.5 and 4.2.6.

4.2.1 Lot. All eductors of the same type and size offered for delivery at one time shall be considered a lot for purposes of conformance inspection.

4.2.2 Sampling for examinations and differential pressure test. A random sample of eductors shall be selected from each lot in accordance with MIL-STD-1916 at Verification Level II for the examinations of 4.2.4 and test of 4.2.5. If the number of nonconforming eductors in any sample exceeds the acceptance number specified in MIL-STD-1916 for the sample, the entire lot shall be tested, and all eductors failing the test shall not be offered for delivery.

4.2.3 Sampling for hydrostatic test. A random sample of eductors shall be selected from each lot in accordance with MIL-STD-1916 at Verification Level II for the test of 4.2.6. If the number of nonconforming eductors in any sample exceeds the acceptance number specified in MIL-STD-1916 for the sample, the entire lot shall be tested, and all eductors failing the test shall not be offered for delivery.



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4.2.4 Examinations.

4.2.4.1 Visual examination. A visual examination shall be performed of the sample eductors selected in accordance with 4.2.2 to verify conformance with the requirements as specified (see table I).

4.2.4.2 Dimensional examination. A dimensional examination shall be performed on the sample eductors selected in accordance with 4.2.2. For Type B eductors, dimensional examination shall verify conformance with 803-8436633.

4.2.4.3 Performance examination. A visual examination of the performance data listed on the nameplates (see 3.3) on the sample eductors selected in accordance with 4.2.2 shall be performed to verify conformance with 3.8.

4.2.5 Differential pressure test. Flat plate strainers shall be subjected to 20 psi differential pressure and checked using a straight edge or other convenient method to ensure there is no permanent distortion.

4.2.6 Hydrostatic test. For Type A eductors, each of the sample eductors selected in accordance with 4.2.3, including the foot valve when applicable, shall be hydrostatically tested at 150 percent of the driving fluid pressure for porosity and strength of casting. For Type B eductors, each of the sample eductors selected in accordance with 4.2.3 shall be hydrostatically tested at the fluid pressures listed in 803-8436633 for the specified components and assembled units. For all eductor types, the pressure shall be applied for a period of 5 minutes, during which time there shall be no leaking or sweating through the joints or casting. If a foot valve is provided, slight leakage around the foot valve seat is acceptable.

TABLE I. Requirements and verifications.

| Description | Requirements | Verifications |
|---|---|---|
| Component parts | 3.1 | 4.2.4.1 |
| Size | 3.2.1 | 4.2.4.2 |
| Nonclogging design | 3.2.2 | 4.2.4.1 |
| Nozzle tip | 3.2.3 | 4.2.4.1 |
| Nozzle access | 3.2.4 | 4.2.4.1 |
| Foot (check) valve and strainer | 3.2.5 | 4.2.4.1 |
| Cylindrical strainers | 3.2.5.1 | 4.2.4.2 |
| Flat plate strainers | 3.2.5.2 | 4.2.5 |
| End connections | 3.2.6 | 4.2.4.1 |
| Marking | 3.3 | 4.2.4.1 |
| Type A eductors | 3.4 | 4.2.4.1 |
| Bolting | 3.4.1 | 4.2.4.1 |
| Connections | 3.4.2 | 4.2.4.1 |
| Type B eductors | 3.5 | 4.2.4.1 |
| Foot valves and strainers | 3.5.1 | 4.2.4.1 |
| Aluminum parts | 3.5.2 | 4.2.4.1 |
| Onboard repair parts | 3.6 | 4.2.4.1 |
| Workmanship | 3.7 | 4.2.4.1, 4.2.6 |
| Operational requirements | 3.8 | 4.2.4.3 |



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## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Eductors conforming to this specification are intended to be used on board naval ships to transfer accumulated water, oil, or gasoline of various types at high volumetric rates under severe conditions for extended periods of time. These eductors are intended to last for the life of the ship on which they are installed.

6.1.1 Type A. Fixed and placed eductor of large size and high volume capability.

6.1.2 Type B. Lightweight and portable design eductor.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type required (see 1.2).
c. Eductor size (determined as being the same as the size of the discharge connection) (see 3.2.1).
d. Applicable flow rate (see 3.2.3).
e. When foot valve and strainer are required (see 3.2.5).
f. Whether cylindrical or flat plate strainer is required (see 3.2.5).
g. Types, sizes, and pertinent details of end connections (see 3.2.6).
h. Connection requirements when other than specified (see 3.4.2).
i. When Type B eductors are required to be other than specified (see 3.5).
j. Operating requirements (see 3.8):
    (1) Fluid or mixture to be handled.
    (2) Expected driving fluid and driving fluid pressure.
    (3) Rated suction lift (self priming).
    (4) Rated discharge head.
    (5) Rated suction capacity.
    (6) Maximum permissible consumption of driving fluid.
k. Packaging requirements (see 5.1).
l. When drawings showing overall dimensions, material, performance data, and weight are required to accompany bids (see 6.3).

6.3 Drawings. When specified (see 6.2), drawings showing overall dimensions, material, performance data, and weight will accompany bids.

6.4 Subject term (key word) listing.

Fluid transfer

Freshwater

Seawater



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6.5 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 4420-2018-001)

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